Date: _____

Submitted By

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Submitted To

Sir Asad

Subject:- PF (programming fundamental)

LAB 1 and 2

PROGRAMS.

Department of CS and IT

1st Semester

Morning.

Hi-Time

Date: _____

PROGRAMS:

TASK 01:

Using characters values:

```
#include <iostream>
using namespace std;
int main()
{
    char letter;
 letter = 'A';
 cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
 cin >> letter;
return 0;
}
```

Output =

```
A
B
```

Hi-Time



```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;
letter = 'A';
    cout << letter << '\n';
       letter = 'B';
    cout << letter << '\n';
 cin >> letter;
  return 0;
}

```

```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;
letter = 'A';
    cout << letter << '\n';
      letter = 'B';
```

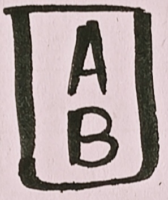
Output

```
A
B
```

Back Copy

PROGRAM 02

```
#include <iostream>
using namespace std;
int main()
{
  char letter;
  letter = 65;
cout << letter << '\n';
  letter = 66;
cout << letter << '\n';
cin >> letter;
return 0;
}
```



Output

```cpp
#include <iostream>
using namespace std;
int main()
{
    int letter;
   letter = 'A';
    cout << letter << '\n';
 letter = 'B';
    cout << letter << '\n';
cin >> letter;
 return 0;
}

```

```cpp
#include <iostream>
using namespace std;
int main()
{
    int letter;
   letter = 'A';
    cout << letter << '\n';
 letter = 'B';
```

Output

```
65
66
```

Back Copy

Task 05

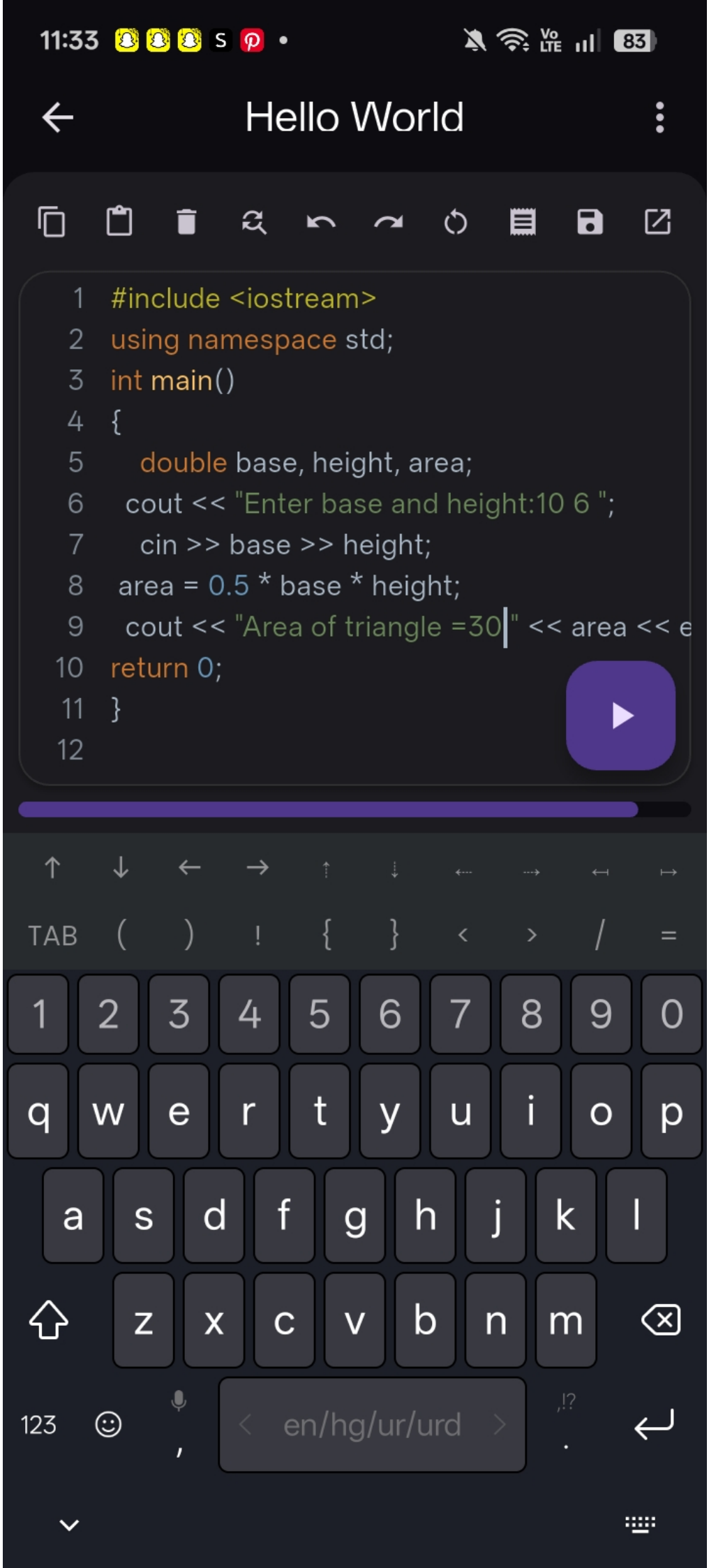
Area of a triangle:

```
#include <iostream>
using namespace std;
int main()
{
    double base, height, area;
    cout << "Enter base and height";
    area = 0.5 * base * height;
    cout << "Area of triangle = " << area << endl;
    return 0;
}
```

base = 10 6

area = 30

output.


11:33
Hello World
#include <iostream>
using namespace std;
int main()
{
double base, height, area;
cout << "Enter base and height:10 6 ";
cin >> base >> height;
area = 0.5 * base * height;
cout << "Area of triangle =30" << area << e
return 0;
}
TAB
en/hg/ur/urd

```
#include <iostream>
using namespace std;
int main()
{
    double base, height, area;
   cout << "Enter base and height:10 6 ";
    cin >> base >> height;
  area = 0.5 * base * height;
                                    << e
```

Output

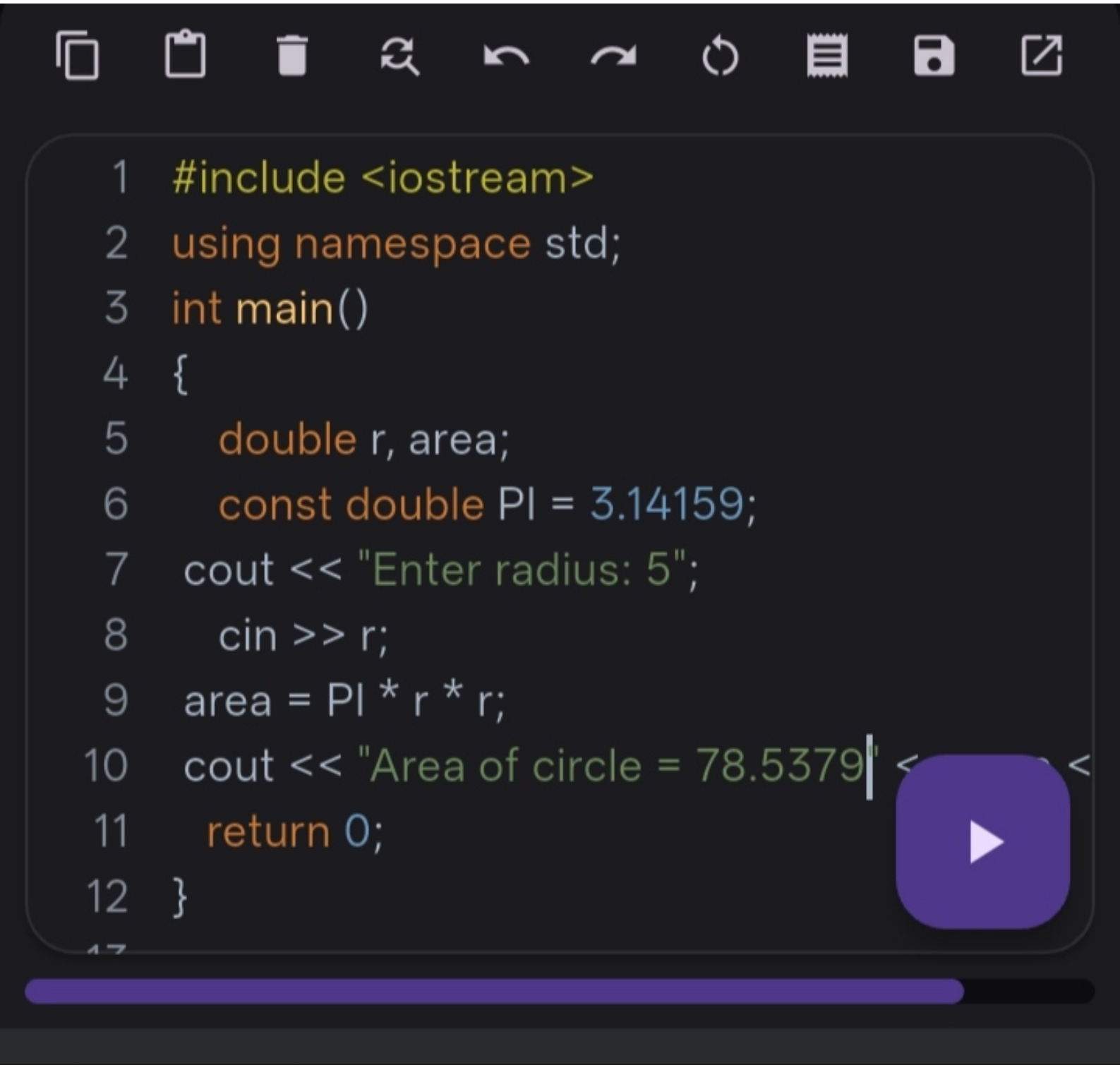
Enter base and height:10 6 Area of triangle =30 0

Back Copy

area = π x radius^2.

```
#include <iostream>
using namespace std;
int main()
{
  double r, area;
  const double PI=3.14159;
  cout << "Enter radius: ";
  cin >> r;
  area = PI * r * r;
cout << "Area of circle=" << area << endl;
  return 0;
}
```

5
78.5397 Output



```cpp
#include <iostream>
using namespace std;
int main()
{
    double r, area;
    const double PI = 3.14159;
 cout << "Enter radius: 5";
    cin >> r;
 area = PI * r * r;
 cout << "Area of circle = 78.5379" <
  return 0;
}
```

```cpp
#include <iostream>
using namespace std;
int main()
{
    double r, area;
    const double PI = 3.14159;
 cout << "Enter radius: 5";
    cin >> r;

rea <
```

Output

```
Enter radius: 5Area of circle =
78.53790
```

Back Copy

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Task 03

Calculate Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
  double x1, y1, x2, y2, d;
  cout << "Enter x1 and y1 : ";
  cin >> x1 >> y1;
  cout << "Enter x2 and y2 ";
  cin >> x2 >> y2;
  d = sqrt(pow(x2-x1,2)+
      pow(y2-y1,2));
  cout << "distance = " << d << endl;
  return 0;
}
```

Output

```
1 2
4 6
5
```

Task 04

Calculate areas of shapes :

Hi-Time